

Top 10 Open-Source Libraries to Fine-Tune LLMs Locally



unisloth



LLaMA-Factory
Easy and Efficient LLM Fine-Tuning



DeepSpeed



Axolotl



PEFT



LitGPT



swift

LLM training and deployment framework



AutoTrain Advanced

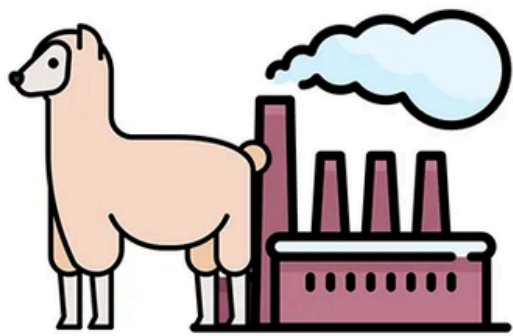


unisloth

Unisloth is built for fast and memory-efficient LLM fine-tuning. It is useful when you want to train models locally, on Colab, Kaggle, or on consumer GPUs. The project says it can train and run hundreds of models faster while using less VRAM.

Best for: Fast local fine-tuning, low-VRAM setups, Hugging Face models, and quick experiments.

Repository: github.com/unislothai/unisloth



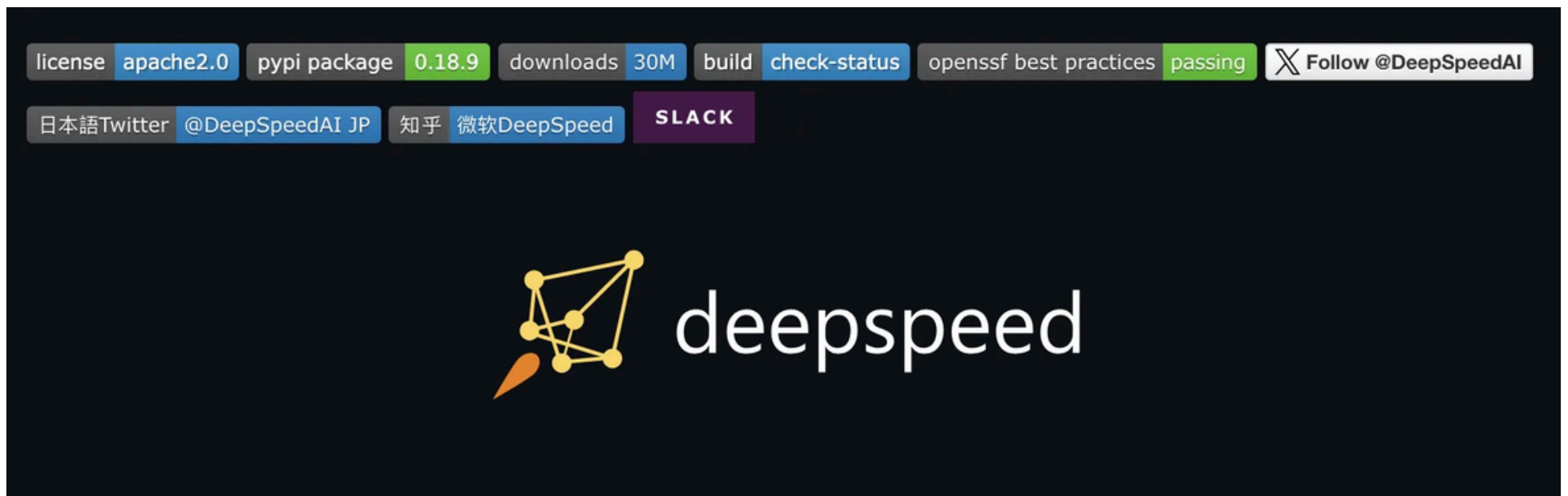
LLaMA-Factory

Easy and Efficient LLM Fine-Tuning

LLaMA-Factory is a fine-tuning framework with both CLI and Web UI support. It is beginner-friendly but still powerful enough for serious experiments across many model families.

Best for: UI-based fine-tuning, quick experiments, and multi-model support.

Repository: github.com/hiyouga/LLaMA-Factory



DeepSpeed is a Microsoft library for large-scale training and inference optimization. It helps reduce memory pressure and improve speed when training large models, especially in distributed GPU setups.

Best for: Large models, multi-GPU training, distributed fine-tuning, and memory optimization.

Repository: github.com/microsoft/DeepSpeed



PEFT stands for Parameter-Efficient Fine-Tuning. It lets you adapt large pretrained models by training only a small number of parameters instead of the full model. It supports methods such as LoRA, adapters, prompt tuning, and prefix tuning.

Best for: LoRA, adapters, prefix tuning, low-cost training, and efficient model adaptation.

Repository: github.com/huggingface/peft



Axolotl

A Free and Open Source LLM Fine-tuning Framework

license Apache-2.0

Tests failing

codecov 4%

release v0.16.1

contributors 237

Stars 12k

discord

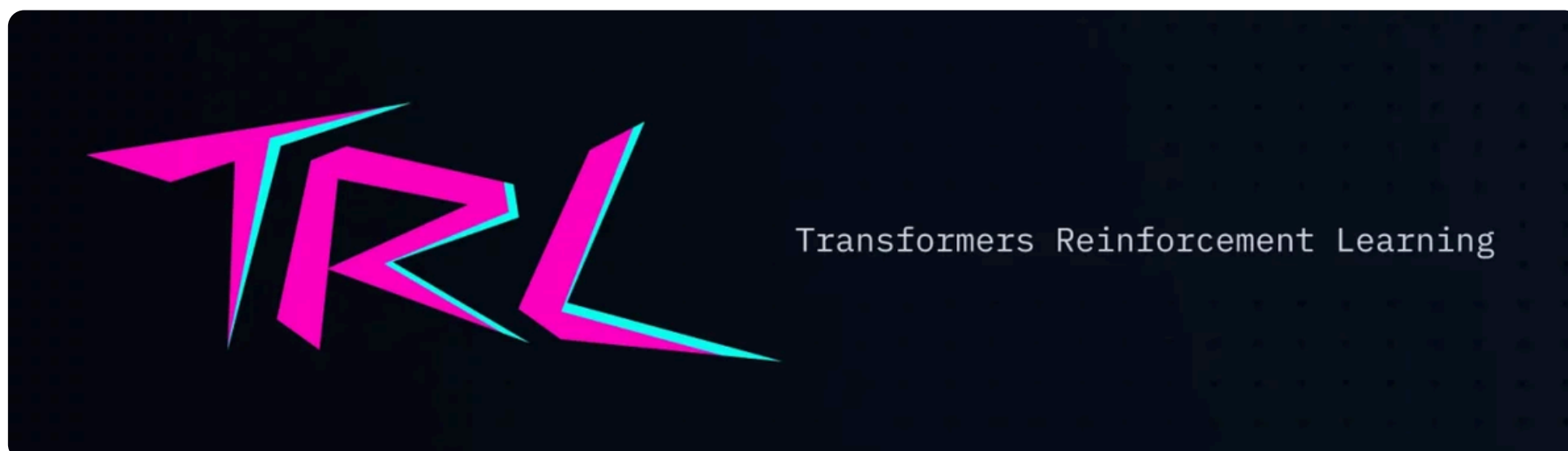
Follow @axolotl_ai

Open in Colab

Axolotl is a flexible fine-tuning framework for users who want more control over the training process. It supports advanced LLM fine-tuning workflows and is popular for LoRA, QLoRA, custom datasets, and repeatable training configurations.

Best for: Custom training pipelines, LoRA/QLoRA, multi-GPU training, and reproducible configs.

Repository: github.com/axolotl-ai-cloud/axolotl



TRL, or Transformer Reinforcement Learning, is Hugging Face's library for post-training and alignment. It supports supervised fine-tuning, DPO, GRPO, reward modeling, and other preference-optimization methods.

Best for: RLHF-style workflows, DPO, PPO, GRPO, SFT, and alignment.

Repository: github.com/huggingface/trl

meta-pytorch/ torchtune



PyTorch native post-training library

165

Contributors

859

Used by

34

Discussions

6k

Stars

720

Forks



torchtune is a PyTorch-native library for post-training and fine-tuning LLMs. It provides modular building blocks and training recipes that work across consumer-grade and professional GPUs.

Best for: PyTorch users, clean training recipes, customization, and research-friendly fine-tuning.

Repository: github.com/meta-pytorch/torchtune

LitGPT

20+ high-performance LLMs with recipes to pretrain, finetune, and deploy at scale.

- | | | |
|-----------------------------------|---------------------|------------------------|
| ✓ From scratch implementations | ✓ No abstractions | ✓ Beginner friendly |
| ✓ Flash attention | ✓ FSDP | ✓ LoRA, QLoRA, Adapter |
| ✓ Reduce GPU memory (fp4/8/16/32) | ✓ 1-1000+ GPUs/TPUs | ✓ 20+ LLMs |

LitGPT provides recipes to pretrain, fine-tune, evaluate, and deploy LLMs. It focuses on simple, hackable implementations and supports LoRA, QLoRA, adapters, quantization, and large-scale training setups.

Best for: Developers who want readable code, from-scratch implementations, and practical training recipes.

Repository: github.com/Lightning-AI/litgpt



SWIFT, from the ModelScope community, is a fine-tuning and deployment framework for large models and multimodal models. It supports pre-training, fine-tuning, human alignment, inference, evaluation, quantization, and deployment across many text and multimodal models.

Best for: Large model fine-tuning, multimodal models, Qwen-style workflows, evaluation, and deployment.

Repository: github.com/modelscope/ms-swift

auto **TRAIN**

Create powerful AI models without code

A new way to automatically train, evaluate and deploy state-of-the-art Machine Learning models.

AutoTrain Advanced is Hugging Face's open-source tool for training models on custom datasets. It can run locally or on cloud machines and works with models available through the Hugging Face Hub.

Best for: No-code or low-code fine-tuning, Hugging Face workflows, custom datasets, and quick model training.

Repository: github.com/huggingface/autotrain-advanced